17. Natural building — durable shelter, stoves, wells, latrines and water

Oracle OS guide. Manuals in the library: "Shelters, Shacks and Shanties", "The Useful Arts Employed in the Construction of Dwelling Houses", "The Science of Brickmaking", "Concrete Construction for the Home and the Farm", "Practical Masonry", Practical Action (rainwater, pumps), WEDC/WHO (sanitation), Appropedia, Open Source Ecology.

Choosing a site

Slightly raised ground (no flooding), near water but latrines more than 30 m from it, sheltered from the prevailing wind, winter sun (south-facing in the northern hemisphere), firewood and good soil nearby.

Wall techniques (simplest to most durable)

- **Log-and-branch hut**: A-frame covered with boughs/foliage/turf — days.
- **Wattle and daub**: stakes and woven rods (willow, hazel) daubed both sides with clay + straw + dung.
- **Cob**: clay + sand + straw trodden together, stacked in 30–50 cm lifts, walls 40–60 cm thick. Great thermal mass.
- **Adobe**: clay + straw blocks moulded and sun-dried 2–4 weeks.
- **Rammed earth**: damp soil tamped between timber forms in 10–15 cm layers.
- **Stone** with lime mortar (see "Basic chemistry"), the most durable; walls wide at the base.
- **Timber frame**: posts and beams with mortise-and-tenon joints (see "Smithing, pottery and carpentry").
- **Golden rule for earth walls**: "**good boots and a hat**" — a stone base above splash height (30–50 cm) and wide eaves so rain never wets the wall.

Roof

Pitch ≥ 45° for thatch (rye/wheat straw, reed) 30 cm thick; fired tiles; slate; overlapping boards; turf over birch bark. Ventilate the roof space.

Insulation and floor

Straw, wool, dry bracken, cork in wall cavities; rammed earth floor over a stone/gravel layer (stops damp) finished with clay + linseed oil.

Stove, chimney and heat

- **Rocket stove**: insulated "L" tube (brick, clay + ash) — wood feeds from the side, flame rises in an insulated chamber; clean burn with little wood.
- **Masonry/thermal-mass stove**: hot smoke runs through a long channel inside a clay/stone bench that stores heat for hours.
- **Chimney** 1 m above the highest point of the roof; sweep once or twice a year (soot fires). Never an indoor fire without a chimney (carbon monoxide).
- **Bread oven** of clay (dome on a stone-and-sand base).

Water

- **Hand-dug well**: only by trained people, with shoring (brick/concrete/timber rings) — collapses and gases kill. Line the top, lid, raised rim, no latrine within 30 m or uphill.
- **Rainwater harvesting**: clean roof → gutter → first-flush diverter → dark covered tank → tap. 1 mm of rain on 1 m² = 1 litre.
- **Slow sand filter** (see Practical Action): 60–90 cm of fine sand over gravel; the top biological layer cleans the water; then disinfect.

Sanitation

- **Pit latrine**: 2–3 m pit, slab with lid, shelter; 30 m from wells and above the water table; ash or lime after each use. When full to 50 cm from the top, cover with soil and dig another.
- **Composting dry toilet**: faeces + sawdust/straw, compost 1–2 years before using on trees (never on raw-eaten vegetables).
- Handwashing next to the latrine (tippy-tap: hanging jerrycan with a rope foot pedal).